Detecting Abnormal Behavior in Autonomous Driving - Prediction of Behavior

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TUM PREP Student Report

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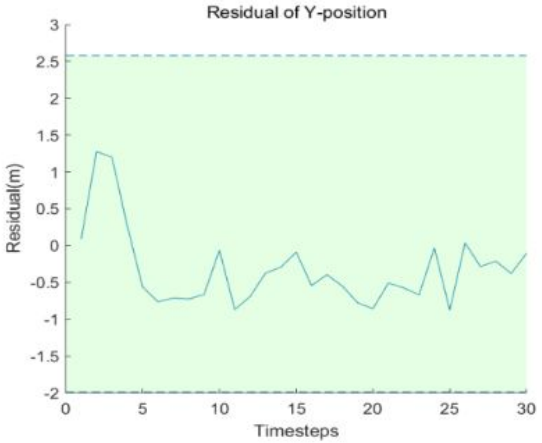
Motivation

- Understand vehicle behavior in traffic
- Expectations are key for safety
- Goal: Predict vehicle direction based on current behavior



Image Courtesy of:
https://www.techsir.com/a/202111/75632.html

Problem Statement



(Wu, 2023)

Report from previous student:

- Extensive estimation conducted
- Thresholds generated
- Validates object movement against expectations

Areas for Improvement:

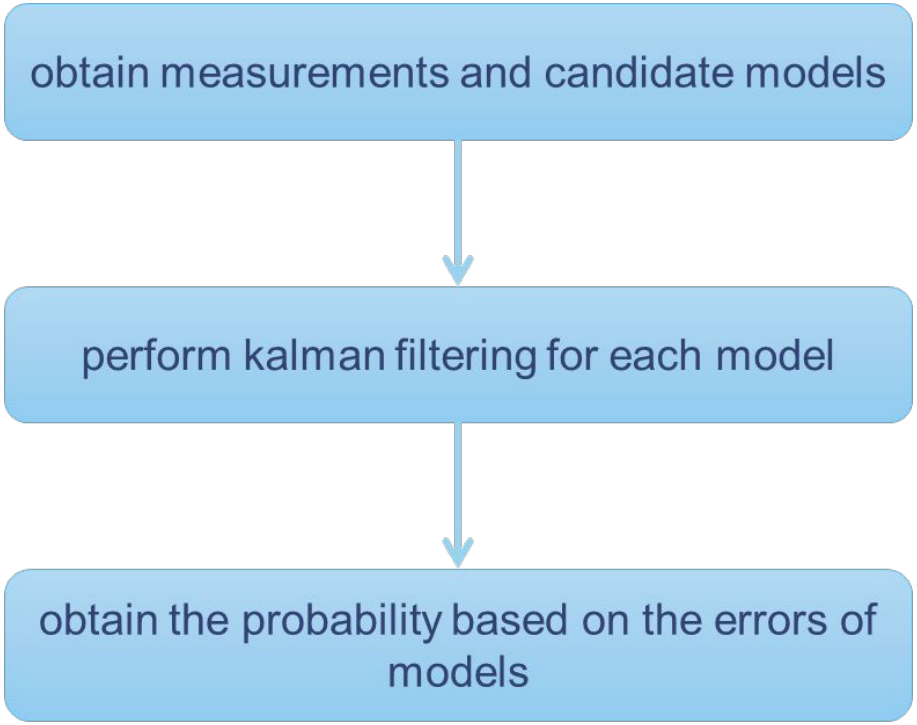
- Thresholds seem inflexible/rigid
- Focuses on just one expectation, limiting efficiency

Our goal this time…

To focus on **more than one model** (possible ways that an object may move) and tell **which maneuver has the highest probability**.

Theory/Methodology

- Use **interacting Multiple Model** (IMM) for object direction probability
- Core of IMM: Application of **Kalman filters***
- One model for each candidate (possible) maneuver



* Kalman filter: a recursive algorithm that optimally **estimates the state of a** linear dynamic **system** from noisy measurements.

Models for Maneuvers

1. State and Control:

$$x = \begin{bmatrix} x_{\text{pos}} \\ v_x \\ y_{\text{pos}} \\ v_y \end{bmatrix} \qquad u = \begin{bmatrix} a_x \\ a_y \end{bmatrix}$$

2. System evolution:

$$x^+ = Fx + Gu$$

3. Control law:

$$u = K(x - x_r)$$

Double integrator:

$$F = \begin{bmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

and

$$G = \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 \\ \Delta t & 0 \\ 0 & \frac{1}{2}\Delta t^2 \\ 0 & \Delta t \end{bmatrix}$$

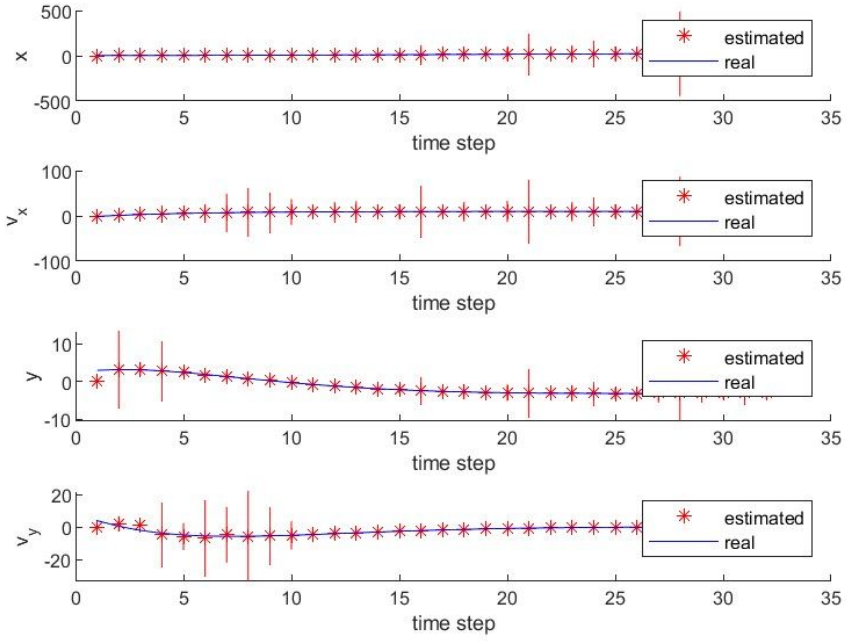
where Δt is the time step.

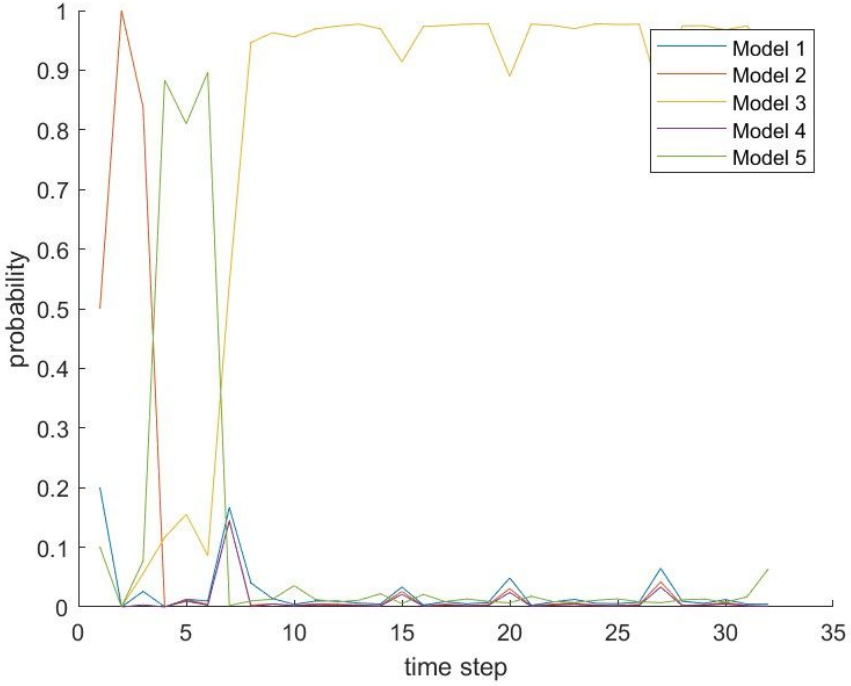
x_r : x_reference, selected maneuver with parameters [x_pos, v_x, y_pos, v_y]

Results

y coordinate of 5 preset models (5 different lanes): 0, 3, -3, 5, -5

Real model with y coordinate **-3** (noise included)

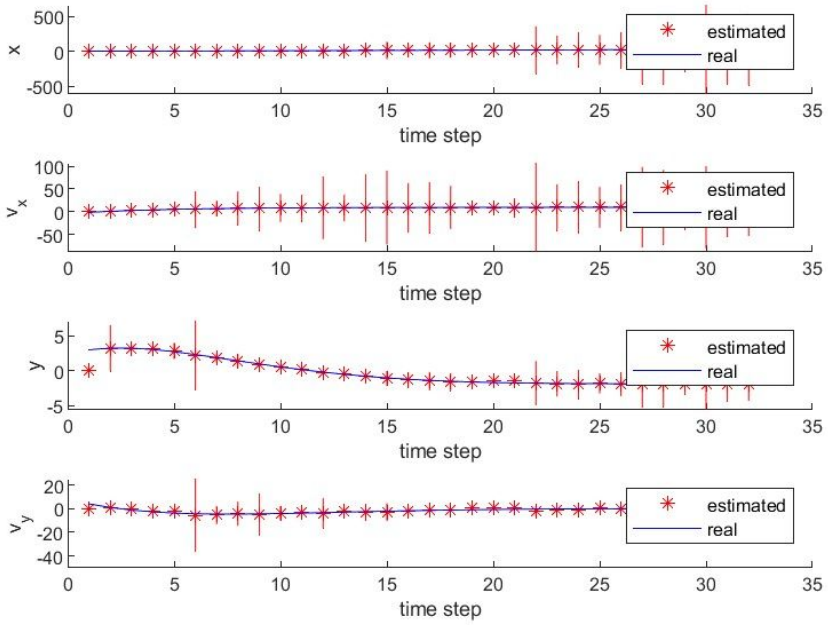


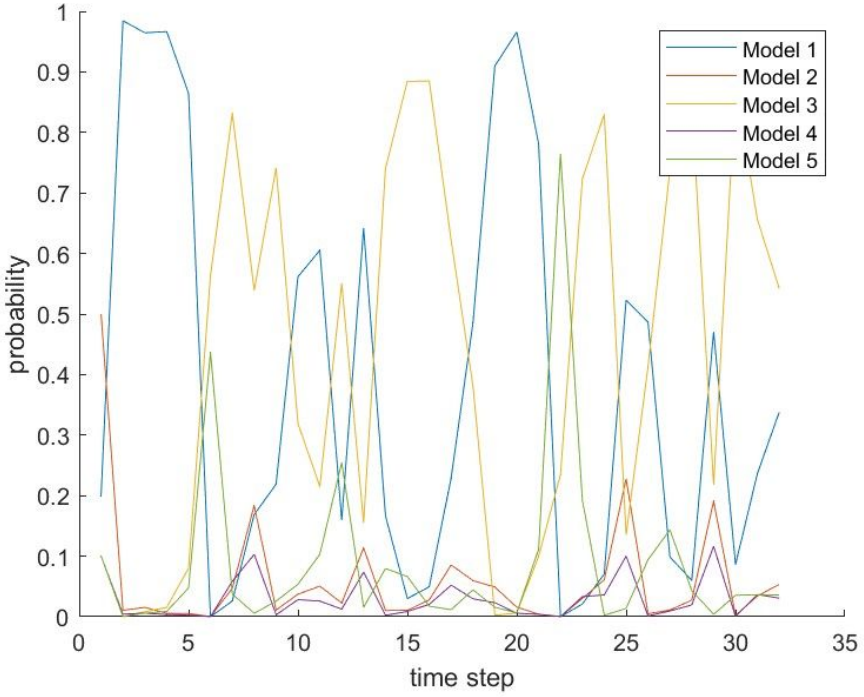


Results

y coordinate of 5 preset models: 0, 3, -3, 5, -5

Real model with y coordinate **-1.75** (noise included)

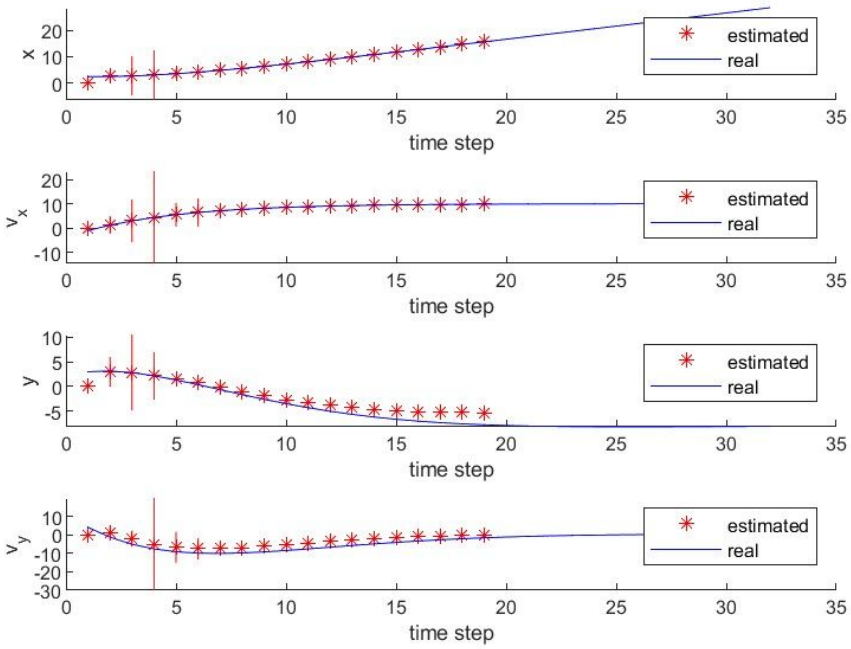


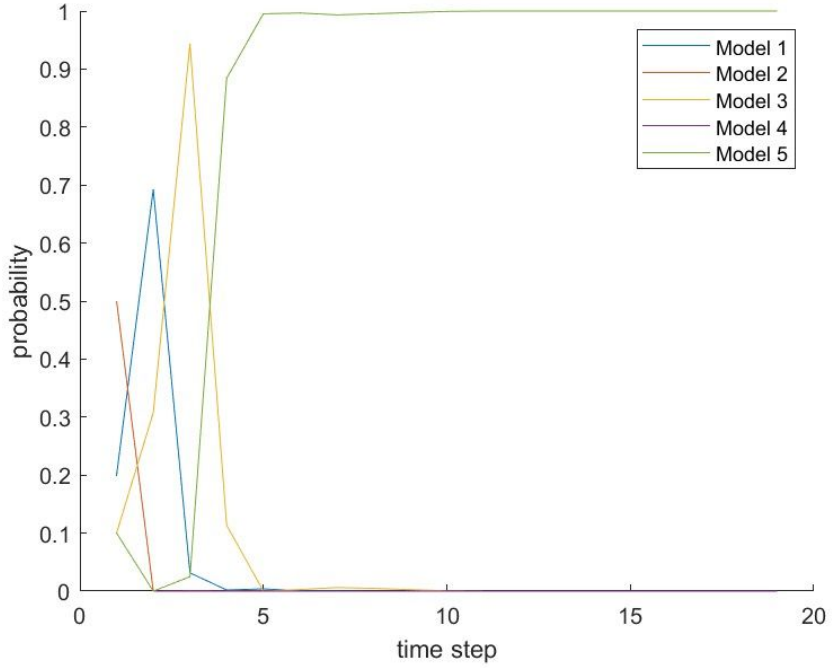


Results

y coordinate of 5 preset models: 0, 3, -3, 5, -5

Real model with y coordinate **-8** (noise included)





Analysis & Conclusion

Summary:

- Design one model for each candidate maneuver
- IMM estimates the most likely correct state and maneuver.

Future Work:

- Increase the number of models tested
- Model parameters should be updated if errors persist

Observations:

- Errors increase as real situations deviate from preset models.
- Plots are less clear when reality aligns with multiple preset models.

References

- Wu, Yu. "Fault Detection for Maneuver Planning in Autonomous Driving." *Department of Electrical and Computer Engineering, Chair of Automatic Control Engineering (LSR), Technical University of Munich, 2023.*